

JINGFEN QIAO

PhD Candidate in Information Retrieval | University of Amsterdam

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RESEARCH INTERESTS

Research Vision: I believe that retrieval is the foundational pillar of Artificial Intelligence, serving as the critical bridge between static knowledge and dynamic reasoning. My research aims to build highly effective, scalable, and robust search systems that empower the next generation of AI through architectural innovations in Agentic Retrieval/Reranking and Efficient Representation Learning.

Core focus areas include **LLM-based Retrieval & Reranking**, **Agentic Search paradigms**, **Learned Sparse Representations**, and **Multimodal Alignment**.

EDUCATION

University of Amsterdam, The Netherlands Oct 2022 – Present

PhD Candidate in Computer Science

Advisors: Prof. dr. Evangelos Kanoulas, Dr. Andrew Yates

Australian National University, Canberra Feb 2019 – Dec 2021

Master of Computing

Relevant Coursework: Document Analysis (NLP) · Machine Learning · Data Mining

Kingston University, London Sep 2013 – Jun 2017

BSc (Hons) Marketing Management

Relevant Coursework: Mathematics for Economists · Economics · Consumer Behaviour · Research Method · Mathematical Analysis

MAJOR PUBLICATIONS

- **Better than Dense? Investigating the Natural Backward Compatibility of Learned Sparse Representations** [PDF] **SIGIR 2026**
Jingfen Qiao, Gabrielle Poerwawinata, Thong Nguyen, Jia-Huei Ju, Eugene Yang, Evangelos Kanoulas, Andrew Yates
- **RE-TRIANGLE: Does TRIANGLE Enable Multimodal Alignment Beyond Cosine Similarity in Retrieval?** [PDF] **SIGIR 2026**
Arijit Ghosh, Aritra Bandyopadhyay, Chiranjeev Bindra, Jingfen Qiao
- **Mitigating Positional Bias in LLM for Listwise Reranking via Calibration and Augmentation** [PDF] **ECIR 2026 (Best Paper)**
Jingfen Qiao, Xinyu Ma, Huang Jin, Dawei Yin, Evangelos Kanoulas, Andrew Yates
- **Reproducibility, Replicability, and Insights into Visual Document Retrieval with Late Interaction** [PDF] **SIGIR 2025**
Jingfen Qiao, Jia-Huei Ju, Xinyu Ma, Evangelos Kanoulas, Andrew Yates
- **DIVAgent: A Diversified Search Agent that Mimics the Human Search Process** [PDF] **CIKM 2025**
Zhirui Deng, Jingfen Qiao, Zhicheng Dou, Ji-Rong Wen, Maarten de Rijke
- **Going Beyond Encoders: Leveraging Decoder Architectures for Learned Sparse Retrieval** [PDF] **KEIR 2025**
Jingfen Qiao, Thong Nguyen, Andrew Yates, Evangelos Kanoulas
- **Intelligent Diagnosis of Northern Corn Leaf Blight with Deep Learning Models** [PDF] **JIA 2022**
Shuaiqun Pan, Jingfen Qiao, Rui Wang, Huilin Yu, Cheng Wang, Kerry Taylor, Hongyu Pan. *Journal of Integrative Agriculture*, 21(4):1094-1105.

WORKING EXPERIENCE

- **Research Intern**, Baidu, Beijing Feb 2024 – Mar 2024
Focused on LLM-based listwise reranking, specifically designing methods for positional bias mitigation and debiasing in generative retrieval systems.
- **Research Assistant**, Computer Network Information Center (CNIC), Chinese Academy of Sciences Feb 2021 – Mar 2022
Developed deep learning models for the digitization of historical archives, with a core focus on Optical Character Recognition (OCR) detection and text recognition.
- **Front-End Developer**, Jemba Australia, Canberra Jul 2020 – Dec 2020
Led the front-end development of a online platform designed to streamline business interactions and service bookings between tradespeople and customers.

TEACHING & MENTORING

- **Student Supervision**, University of Amsterdam 2025 – Present
Supervised and co-supervised the following master's theses and research projects:
 - **Robbert Sierhuis**: *Retrieving Scientific Content with Vision–Language Models: A Domain-Specific Evaluation* (2025)
 - **Ioannis Kapetangeorgis**: *Multi-View Image Retrieval* (2026)
 - **Arijit Ghosh, Aritra Bandyopadhyay, Chiranjeev Bindra**: *RE-TRIANGLE: Does TRIANGLE Enable Multimodal Alignment Beyond Cosine Similarity in Retrieval?* (2026)
- **Teaching Assistant**, Information Retrieval 2, University of Amsterdam 2023 – 2025
- **Teaching Assistant**, Data Wrangling, Australian National University Feb 2020 – May 2020
Supported students in data pre-processing and mining techniques using Python.
- **Teaching Assistant**, Web Development, Kingston University 2014 – 2016
Assisted in teaching front-end development and brand management principles.

ACADEMIC SERVICE

- **Reviewer /Sub-reviewer:**
ACM SIGIR (2025, 2026) · ACL (2025) · ECIR (2024, 2025)
- **Journal Reviewer:**
Information Processing & Management

PRESENTATIONS & TALKS

- **Conference Talks** *Oral/Poster Presentations* · SIGIR (2025) · ECIR (2025, 2026)
- **Invited Talk at eBay** *Invited Talk* · Presented *DebiasFirst* research paper (2026)
- **Invited Talk at SEA** *Invited Talk* · Presented work on *Visual Document Retrieval* (2025)
- **Workshop Presentations** *Lightning Talk* · Search Futures (2026) · *Poster* · KEIR (2025)

AWARDS & RECOGNITION

- **Best Paper Award**, European Conference on Information Retrieval (ECIR) 2026
- Kingston University Scholarship 2013 - 2016
- First prize in Citibank Young Entrepreneur Program 2012

TECHNICAL SKILLS

- **Programming Languages:** Python · Java · SQL · PHP
- **Machine Learning & NLP:** PyTorch · TensorFlow · vLLM · PEFT (LoRA) · LangChain · LlamaIndex